DEPARTMENT OF WORKS & HIGHWAYS

MATERIAL SCIENCE & TECHNOLOGY BRANCH

Materials Testing Laboratory

[√] **Port Moresby,** [ ] **Mt. Hagen,** [ ] **Lae,** [ ] **Annex Lab.**

Date:

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AGGREGATE CRUSHING VALUE - AS 1141. 21

Client : ---------- Asset No. ----------

Project: ---------- Job No: ----------

Location: ---------- Material Source / Pit No. ----------

Material Description: ---------- Depth: ----------

Nominal Size of Aggregate _____ mm

1. Standard size aggregate Passing 13.2mm sieve and retianed on 9.5mm sieve
2. Other size aggregate Passing ---------- mm sieve and retained -------- mm sieve

Test Portion Number			**X**	**Y**
C	Mass of sample + measure before crushing	g		
M	Mass of measure	g		
A	Mass of sample (C-M)	g		
L	Maximum load applied	kN		
D	Mass of fraction retained on 2.36mm sieve after crushing	g		
B	Mass of fraction passing 2.36mm sieve (A - D)	g		
ACV	Aggregate Crushing Value (B/A) * 100	%		
	Average ACV (X+Y)/2	%		

TEST CONDITIONS FOR VARIOUS AGGREGATE SIZES

Nominal Size of Aggregate (mm)		Sieve sizes for seperating fines(mm)	Quantity of Material required (kg)	Required total force (kN)	Diameter of Cylinders used (kN)
Passing	Retained				
53	37.5	9.5	75	1600	300
37.5	26.5	6.7	75	1600	300
26.5	19.0	4.75	75	400	150
19.0	13.2	3.35	10	400	150
13.2	9.5	2.56	10	400	150
9.5	6.7	1.70	10	400	150
6.7	4.75	1.18	1.2	100	100
4.75	3.35	850 micron	1.2	100	100
3.35	2.36	600 micron	1.2	100	100

Tested by: ---------- Checked by: ----------

Date: ---------- Date: ----------

Authorised by: AS(Material Science & Technology Branch)

FORM No. A-21

Revised Worksheet:: 09/05/2022 lhb